Local surrogates for quantum machine learning

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SCAN ME

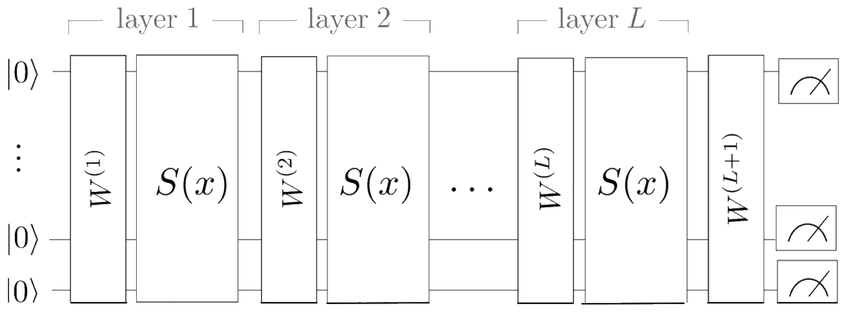
Reuploading models

A univariate quantum model f(x) defined as the expectation value of some observable with respect to a state prepared through variational quantum circuit with parameters θ.

$$f_\theta(x) = \langle 0|U^\dagger(x,\theta)MU(x,\theta)|0\rangle$$

As it consists of L layers each with a data encoding block and a trainable block it is refered to as re-uploading type models. These can be written as a partial Fourier series (Schuld et al (2021)):

$$f(x) = \sum_{n\in\omega} c_n e^{inx}$$

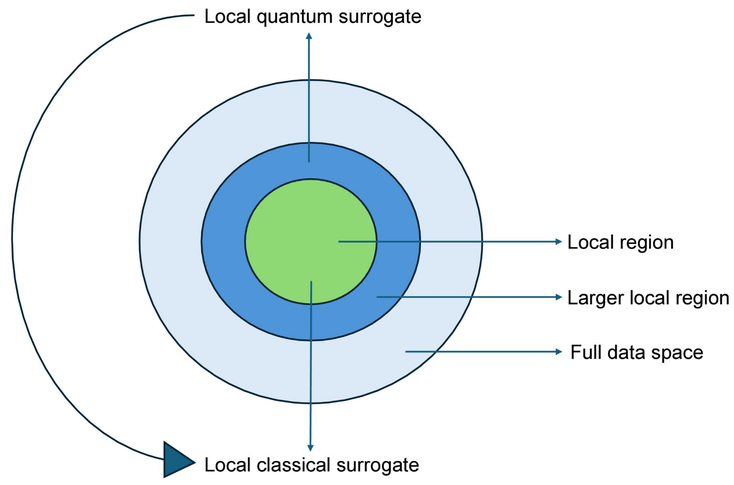


Local Surrogates

Proposal:
Local surrogates for quantum learning models and Local surrogation protocol (LSP)

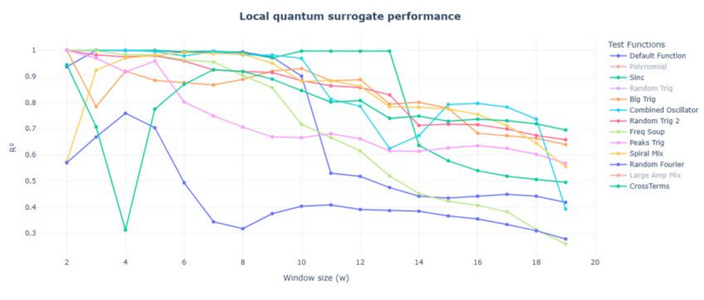
DEF: A hypothesis class of quantum learning models F has quantum/classical surrogates if there exists a mapping $S : F \to G$ such that for each $f \in F$, the corresponding quantum/classical model $g_f \in G$, satisfies that the maximal deviation $|f(x) - g_f(x)|$, is bounded with high probability within a local domain $x \in \mathcal{X}_{\text{window}}$

$$\mathbb{P}\left[\sup_{x\in\mathcal{X}_{\text{window}}} \|f(x) - g_f(x)\| \leq \epsilon\right] \geq 1 - \delta$$

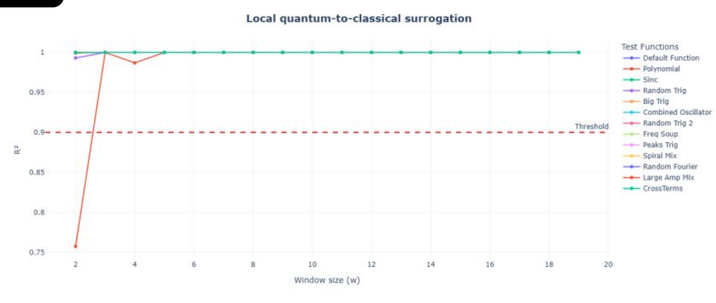


Local surrogation. For a small enough local region within the data space of a quantum learning model, a cost-efficient local quantum/classical surrogate can be generated with adequate accuracy, potentially as a two-step protocol.

LSP on 2D functions



(a) Local quantum surrogate fit to target function (window size vs R^2)



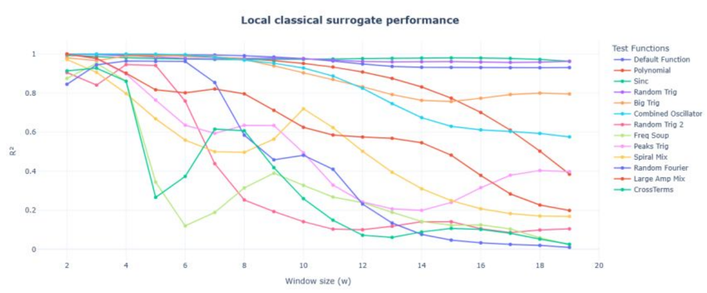
(b) Local classical surrogate fit to the local quantum surrogate, wherein the number of reuploading layers is fed to the local classical surrogate. (window size vs R^2)

Global Surrogates

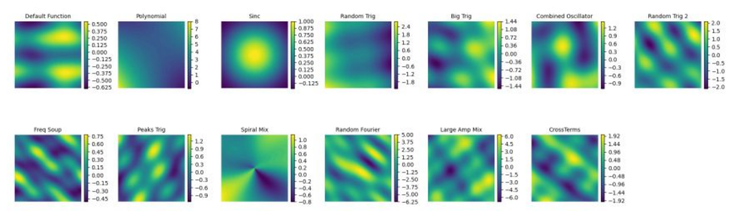
DEF: A hypothesis class of quantum learning models F has classical surrogates if there exists a mapping $S : F \to G$ such that for each $f \in F$, the corresponding classical model , $g_f \in G$ satisfies that the maximal deviation , $|f(x) - g_f(x)|$ is bounded with high probability

$$P\left[\sup_{x\in\mathcal{X}} \|f(x) - g_f(x)\| \leq \epsilon\right] \geq 1 - \delta$$

Fourier based surrogates:
Truncated fourier representation of re-uploading type quantum learning models admit construction of classical Fourier based surrogates for that particular category of learning models. (Schreiber et al (2022))
The number of layers in the re-uploading type model determines the required frequency spectrum for the fourier surrogates.

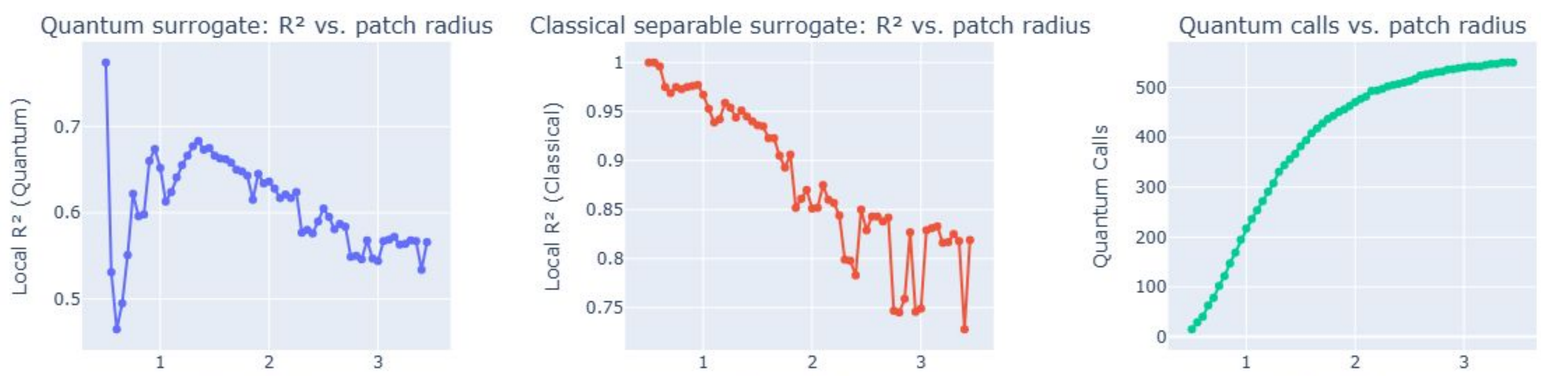


(c) Local classical surrogate fit to target function with the latter treated as a black-box model (window size vs R^2)



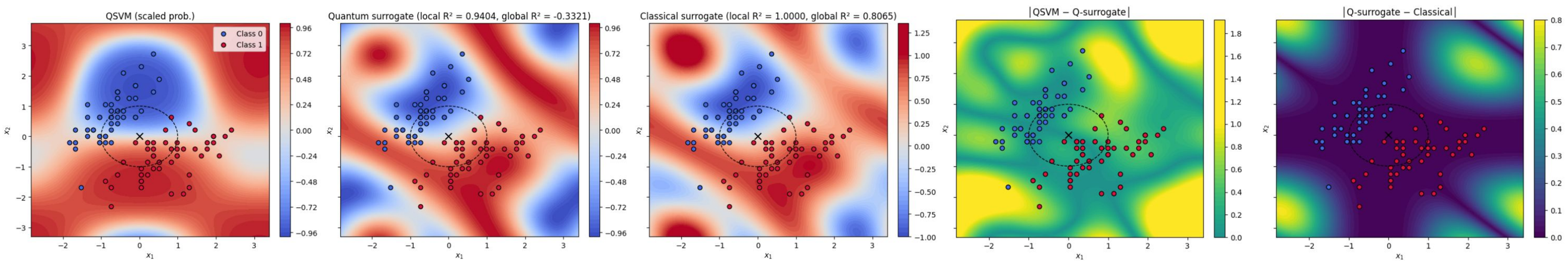
(d) 2-D target functions used in the experiment (definitions provided in Table I).

LSP on Cancer dataset



Des: Local surrogation protocol (LSP) on PCA reduced 6-feature Breast Cancer Wisconsin (Diagnostic) data set to test the practical limits of classical surrogation protocol. Hypercube patch radii ranged between 0.5 - 3.5 with step size 0.05. We design the 2-qubit 3-layer reuploading local quantum surrogate.

LSP on QSVM (IRIS)



(a) Local and global performances of local quantum surrogate trained on sampled data from the QSVM and local classical surrogate trained on sampled data from local quantum surrogate.

(b) Relative errors of local quantum surrogate against QSVM and local classical surrogate against local quantum surrogate.

Des: Local surrogation protocol (LSP) on a QSVM trained on a 2-class, 2-feature Iris data set. QSVM had a R2 of 0.84 after its training. PennyLane was used to design the 2-qubit QSVM and the 1-qubit 2-layer reuploading local quantum surrogate. The local quantum-to-classical surrogate is implemented as per the classical surrogation protocol, with the local quantum surrogate treated as a white-box model.